

Pranav Rathod

pranavdev@duck.com | PranavRathod.com | LinkedIn.com/in/PranavSRathod

EDUCATION

University of Southern California

Master of Science in Computer Science, Multimedia and Creative Technologies

January 2024 – December 2025

Los Angeles, CA

University of Illinois at Chicago

Bachelor of Science in Computer Science, Minor in Art

August 2019 – May 2023

Chicago, IL

RELEVANT COURSEWORK

Analysis of Algorithms, Multimedia Systems Design, Computer Graphics, Robotics, Computer Animation and Simulation, Game Engine Development, 3D Graphics and Rendering, Artificial Intelligence, Data Structures, Software Design, Database Systems

SKILLS

Languages: C/C++, Python, C#, Java, JavaScript, SQL, Dart, Go, F#

Technical: OOP, Git, CI/CD (GitHub Actions), Docker, Terraform, Linux/Unix, OpenGL, PyTorch, Node.js, React, HTML5, CSS3, Unity, WebGL, Firebase, GCP, SQLite3, Multi-Threading, TCP/IP, Testing Frameworks (Google/JUnit5)

EXPERIENCE

Media Support Specialist

Annenberg TechOps Information Technology, University of Southern California

January 2024 – December 2025

Los Angeles, CA

- Troubleshoot cross-platform rendering and asset pipeline issues, ensuring reliable deployment of multimedia applications
- Resolve web and multimedia workflow issues by troubleshooting HTML, WordPress, and Adobe Creative Cloud
- Authored technical documentation and tutorials to streamline multimedia production for 200+ students and faculty

Research Assistant

ELiCIT Lab, University of Illinois at Chicago

June 2022 – December 2023

Chicago, IL

- Collaborated with Adobe Research to prototype a Flutter-based mobile app for multimodal interaction studies
- Explored backend trade-offs in storing synchronized multimodal data for use in training NUI models
- Developed UI workflows and backend logic to manage speech and gesture-driven image editing tasks across 12 workflows
- Integrated GCP Speech-to-Text API and installed real-time gesture logging using touch coordinates and timing data

Teaching Assistant

Department of Computer Science, University of Illinois at Chicago

January 2022 – December 2022

Chicago, IL

- Led community discussions and labs covering recursion, graph traversal, and dynamic programming for 100+ students
- Guided students in debugging and writing code in C, C++ and building applications using Flutter, Dart across 3 semesters
- Formulated course logistics and graded assignments for intermediate and advanced Computer Science courses each semester

PROJECTS

Autonomous Robotic Navigation & Manipulation Pipeline | Python, ROS2, NumPy

December 2025

- Developed a particle-filter localization with 3.5k particles, maintaining stable pose estimates under severe sensor noise
- Reduced manipulator path length by 30% via RRT shortcut-based smoothing while preserving collision safety
- Executed precision grasping with sub-centimeter accuracy via Jacobian control and tuned integration step sizes

PrimeEngine | C++, OpenGL, Maya

April 2025

- Built a scalable real-time 2D overlay and player controls to enhance debugging efficiency in a custom game engine
- Optimized rendering performance by 40% via algorithmic AABB frustum culling and integrated real-time visualization tools
- Integrated Maya asset workflow and optimized render loop to support faster design-to-engine iteration cycles

Motion Capture Interpolation | C++, OpenGL, FLTK

March 2025

- Reconstructed missing motion capture sequences using Linear, Bezier, SLERP, and Bezier SLERP interpolation methods
- Evaluated performance versus accuracy trade-offs in interpolation methods under real-time compute and memory constraints
- Implemented OOP-based playback system with FLTK UI to compare interpolation quality in real time

Segmented Video Compression | Python, Detectron2, OpenCV, PyTorch

December 2024

- Engineered block-based encoder using DCT, quantization, and motion estimation to compress .rgb videos efficiently
- Achieved 95% accuracy in foreground-background separation with Detectron2 and optical flow for adaptive compression
- Built synchronized A/V player supporting 30 FPS playback with real-time IDCT decoding and pause/step controls

ACHIEVEMENTS & LEADERSHIP

SIGGRAPH Student Volunteer Team Leader | Denver, CO; Tokyo, Japan; Vancouver, BC

2024–2025

- Oversaw operations for 350+ international student volunteers at world's premier computer graphics conference
- Facilitated event logistics and attendee support, collaborating with global teams of students and industry professionals

Undergraduate Research Forum | Chicago, IL

April 2023

- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research

Engineering Expo | Chicago, IL

April 2021

- Won 'Best in Category' for designing an Arduino-based device to enforce COVID-19 social distancing